

FINAL PROJECT REPORT

SMARTPHONE USAGE, DIGITAL ADDICTION AND WELL-BEING ANALYTICS

An Interactive Data Analytics and Visualization Project using Microsoft Power BI

Submitted by: MUNNANGI PRISKILLA

Team ID / Roll No: 2025MCA16059

Course: Master of Computer Applications (MCA)

University: Sri Padhmavathi Mahila Visvavidhyalayam, Tirupati

Academic Year: 2026–2027

Project Domain	Data Analytics / Business Intelligence
Primary Tool	Microsoft Power BI
Dataset	Smartphone Usage Dataset – 7,500 user records and 16 columns
Project Focus	Smartphone usage behaviour, digital addiction and digital well-being
Team ID	MCA

Table of content

- 1. Introduction
 - 1.1. Project overviews
 - 1.2. Objectives
- 2. Project Initialization and Planning Phase
 - 2.1. Define Problem Statement
 - 2.2. Project Proposal (Proposed Solution)
 - 2.3. Initial Project Planning
- 3. Data Collection and Preprocessing Phase
 - 3.1. Data Collection Plan and Raw Data Sources Identified
 - 3.2. Data Quality Report
 - 3.3. Data Exploration and Preprocessing
- 4. Data Visualization
 - 4.1. Framing Business Questions
 - 4.2. Developing Visualizations
- 5. Dashboard
 - 5.1. Dashboard Design Screenshot with Insights
- 6. Report
 - 6.1. Report Design Screenshot with Insights
- 7. Performance Testing
 - 7.1 Utilization of Data filters
 - 7.2 No of Calculation Columns/Measures
 - 7.3 No of Visualization
- 8. Conclusion/Observation
- 9. Future Scope
- 10. Appendix
 - 10.1. Source Code(if any)
 - 10.2. GitHub & Project Demo Link

1. Introduction

1.1 Project Overview

Project Title: ***Smartphone Usage, Digital Addiction & Well-Being Analytics***

This project focuses on analyzing smartphone usage patterns, digital addiction, and digital well-being using Microsoft Power BI.

The project uses a Smartphone Usage dataset containing 7,500 user records and 16 columns. The dataset includes demographic information and smartphone usage factors such as daily screen time, weekend screen time, social media usage, gaming, sleep, stress, notifications, app openings, work/study hours, academic/work impact, and addiction information.

Power Query is used for data preprocessing, while Power BI is used for data modelling, analysis, visualization, and dashboard development.

1.2 Objectives

- Quantify smartphone addiction among the 7,500 users.

Analyze addiction across age groups, gender, and stress levels.

- Analyze factors associated with addiction, including screen time, social media, notifications, sleep, and stress.
- Analyze addiction severity using None, Mild, Moderate, and Severe categories.
- Create an Age Group field for segment-level analysis.
- Develop an interactive Power BI dashboard.
- Provide data-driven insights into smartphone usage and digital well-being.

2. Project Initialization and Planning Phase

2.1 Define Problem Statement

The project addresses the difficulty of understanding smartphone usage and digital addiction patterns from raw data.

Problem Statement 1:

A smartphone user wants to understand daily smartphone usage, screen time, social media and gaming habits, but different usage measurements such as screen time, app usage and notifications are difficult to interpret without proper analysis.

Problem Statement 2:

A student wants to understand how smartphone usage patterns relate to sleep, stress and academic/work activities, but these measurements are available separately and are not presented in an easy-to-understand analytical form.

The project therefore focuses on converting raw smartphone usage data into clear and interactive analytical insights.

2.2 Project Proposal – Proposed Solution

The proposed solution is an **interactive Smartphone Addiction and Digital Well-Being Analysis Dashboard using Microsoft Power BI**.

The solution:

- Analyzes 7,500 users.
- Uses Power Query for data preparation.
- Uses Power BI for analysis and visualization.
- Provides KPI cards and interactive charts.
- Analyzes addiction levels, screen time, sleep, stress, social media, gaming, app opens, notifications and work/study hours.
- Provides filters/slicers and Key Influencers analysis.

2.3 Initial Project Planning

The project was planned through four major sprints:

Sprint 1 – Data Collection & Understanding

- Collect the Smartphone Usage dataset.
- Understand columns, records and data types.
- Identify important attributes.
- Check and handle missing, duplicate and inconsistent data.

Sprint 2 – Transformation & Analysis

- Transform and prepare the data.
- Analyze daily screen time, social media, gaming, app opens and notifications.

- Analyze addiction levels and usage patterns.

Sprint 3 – Well-Being & Demographic Analysis

- Analyze sleep, stress and academic/work activities.
- Compare usage patterns across age groups and genders.
- Develop the Power BI dashboard.

Sprint 4 – Dashboard, Insights & Report

- Add dashboard filters.
- Generate important insights.
- Document analysis, visualizations and conclusions in the final report.

3. Data Collection and Preprocessing Phase

3.1 Data Collection Plan and Raw Data Sources Identified

Dataset: Smartphone_Usage.csv

- **Records:** 7,500
- **Columns:** 16
- **Format:** CSV
- **Source:** Project-provided local dataset

Important Dataset Fields

- Transaction ID / User ID
- Age
- Gender
- Daily Screen Time
- Weekend Screen Time
- Social Media Hours
- Gaming Hours
- Work/Study Hours
- Sleep Hours
- Notifications

- App Opens
- Stress Level
- Academic/Work Impact
- Addiction Level
- Addicted Label

The processed dataset is used in Power BI for KPIs, analysis and dashboard visualizations.

3.2 Data Quality Report

The dataset was checked for the following quality issues:

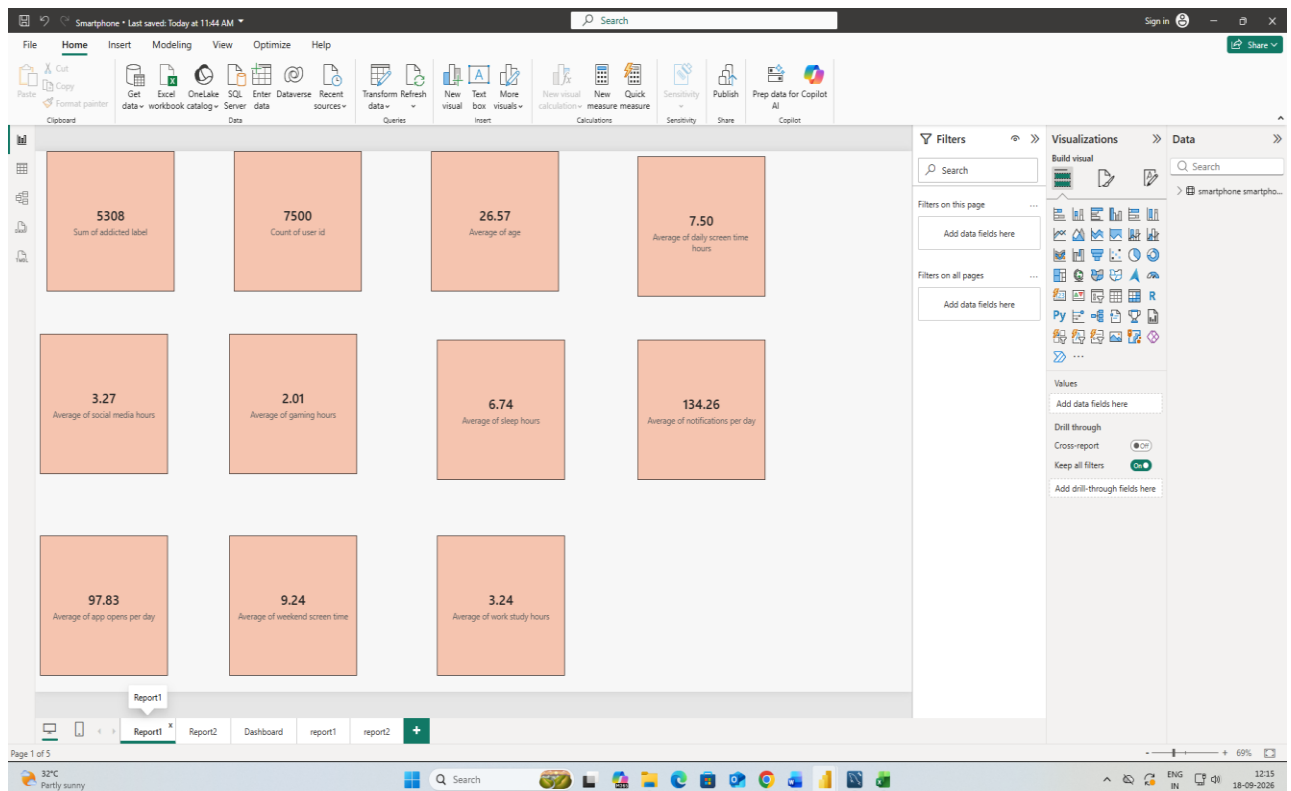
- Missing or blank values
- Duplicate records
- Inconsistent category names
- Incorrect data types
- Different age values that are difficult to compare directly
- Inconsistent addiction categories
- Invalid numerical values
- Formatting inconsistencies

Resolution

Power Query was used to:

- Identify and handle missing values.
- Check and remove duplicate records where required.
- Standardize Gender, Stress Level and Addiction Level categories.
- Convert numerical fields into suitable data types.
- Check numerical values such as screen time, sleep, gaming and social media hours.
- Standardize addiction categories as **None, Mild, Moderate and Severe**.
- Apply consistent formatting.

3.3 Data Exploration and Preprocessing



Data Transformation

Power Query was used for filtering, sorting and preparing the data.

Age Group Creation

An **Age Group** calculated field was created with:

- **Under 20**
- **20–29**
- **30–39**

Data Type Conversion

- Screen time, sleep, gaming and social-media hours → numerical fields.
- Gender, stress level and addiction level → categorical fields.

Data Modelling

The processed data was loaded into Power BI and measures were created for:

- User count
- Average screen time

- Average sleep hours
- Average social-media hours
- Average gaming hours
- Average app opens
- Average notifications

The processed data was then used to create the final Power BI dashboard.

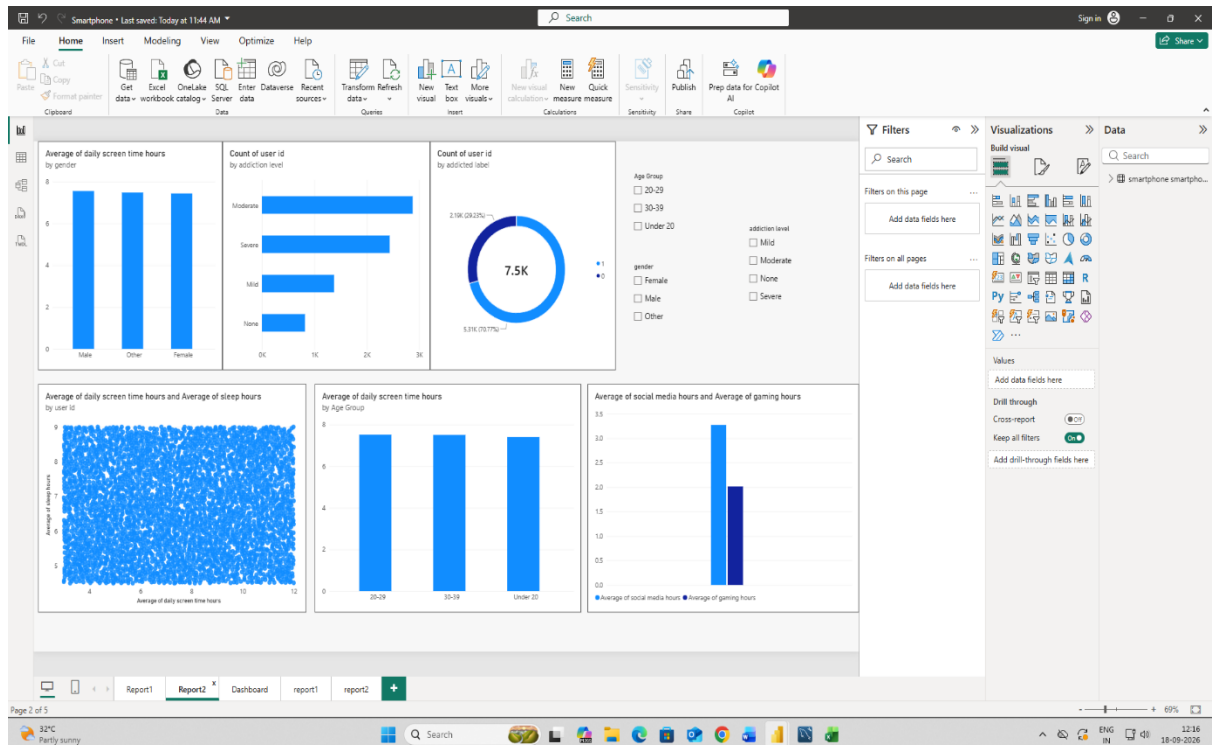
4. Data Visualization

4.1 Framing Business Questions

The project answers the following business questions:

1. What percentage of users are classified as addicted?
2. How many users belong to each addiction level?
3. How does average daily screen time vary by gender?
4. How does screen time vary across age groups?
5. What is the relationship between daily screen time and sleep hours?
6. How do gaming and social-media usage compare?
7. What are the key smartphone usage and well-being indicators?
8. What is the average age and total user population?
9. How do gaming and social-media usage vary by addiction level?
10. How does weekend screen time vary by addiction level?

4.2 Developing Visualizations

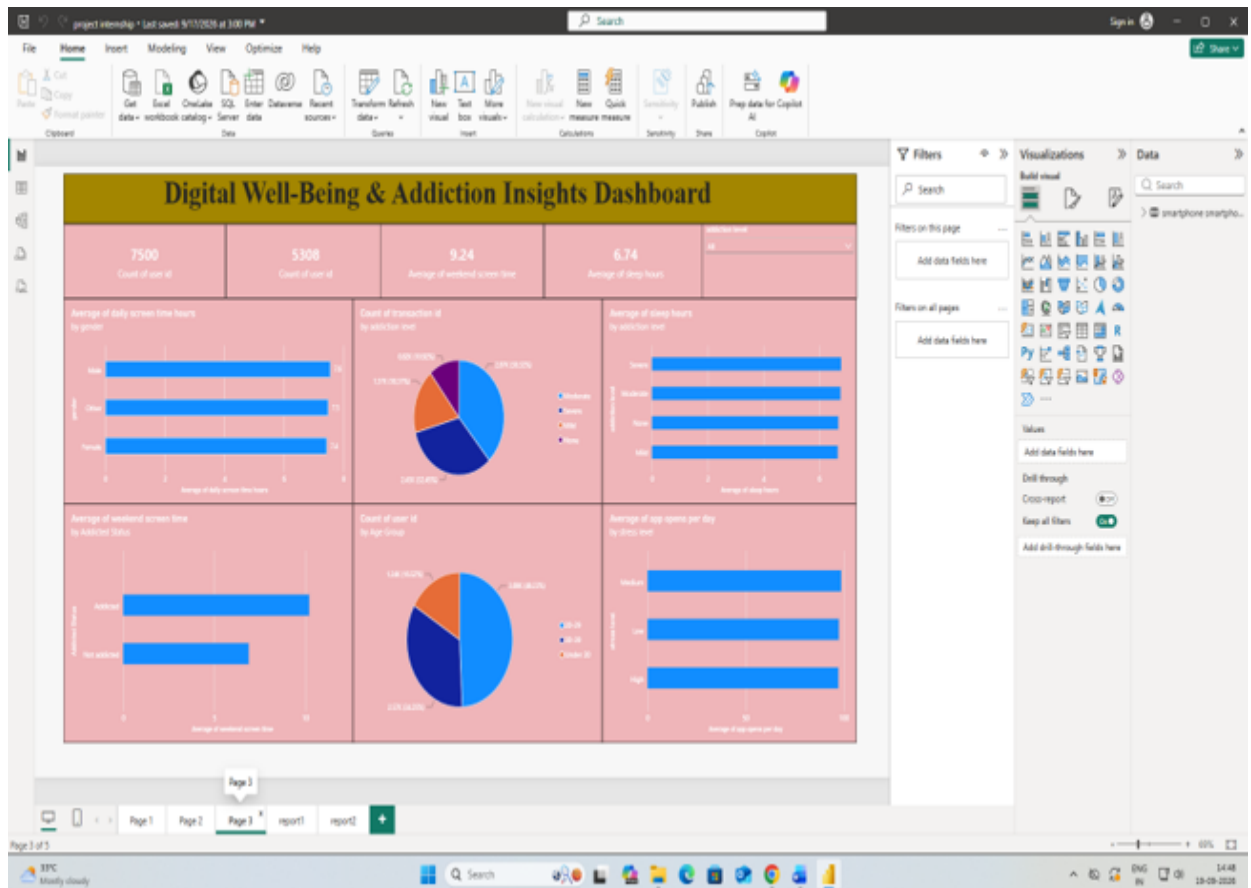


The project uses:

- KPI Cards
- Donut Charts
- Bar Charts
- Column Charts
- Scatter Plot
- Line Chart
- Decomposition Tree
- Key Influencers
- Interactive Filters and Slicers

5. Dashboard

5.1 Dashboard Design Screenshot with Insights



Major Dashboard Outcomes

- **Total Users:** 7,500
- **Average Daily Screen Time:** 7.50 hours
- **Average Weekend Screen Time:** 9.24 hours
- **Average Sleep:** 6.74 hours
- **Average App Opens:** 97.83/day
- **Average Notifications:** 134.26/day
- **Average Social Media Usage:** 3.27 hours
- **Average Gaming Usage:** 2.01 hours
- **Average Work/Study Time:** 3.24 hours
- **Average Age:** 26.57 years

Addiction Distribution

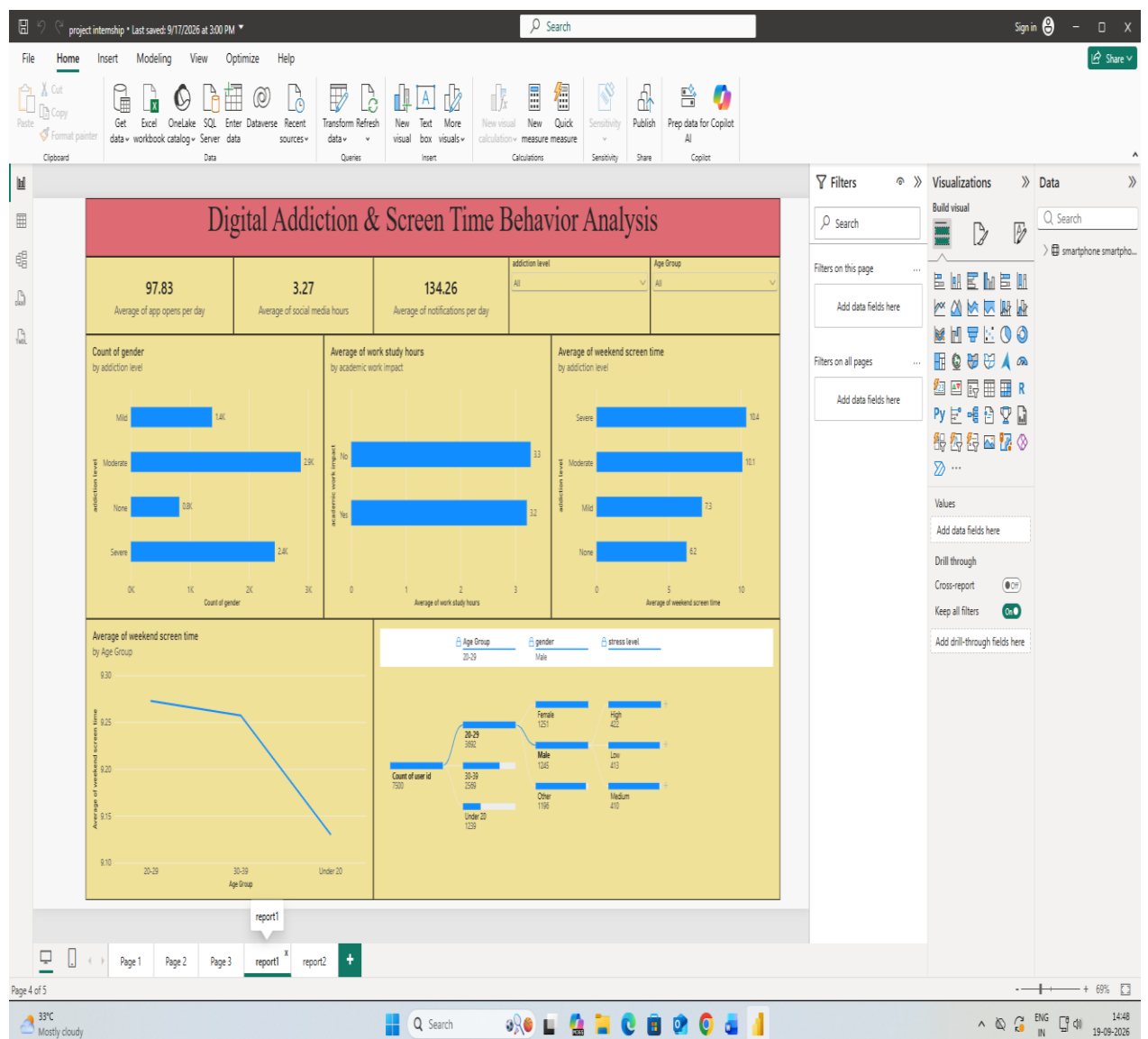
- **Moderate:** 2.87K (38.32%)

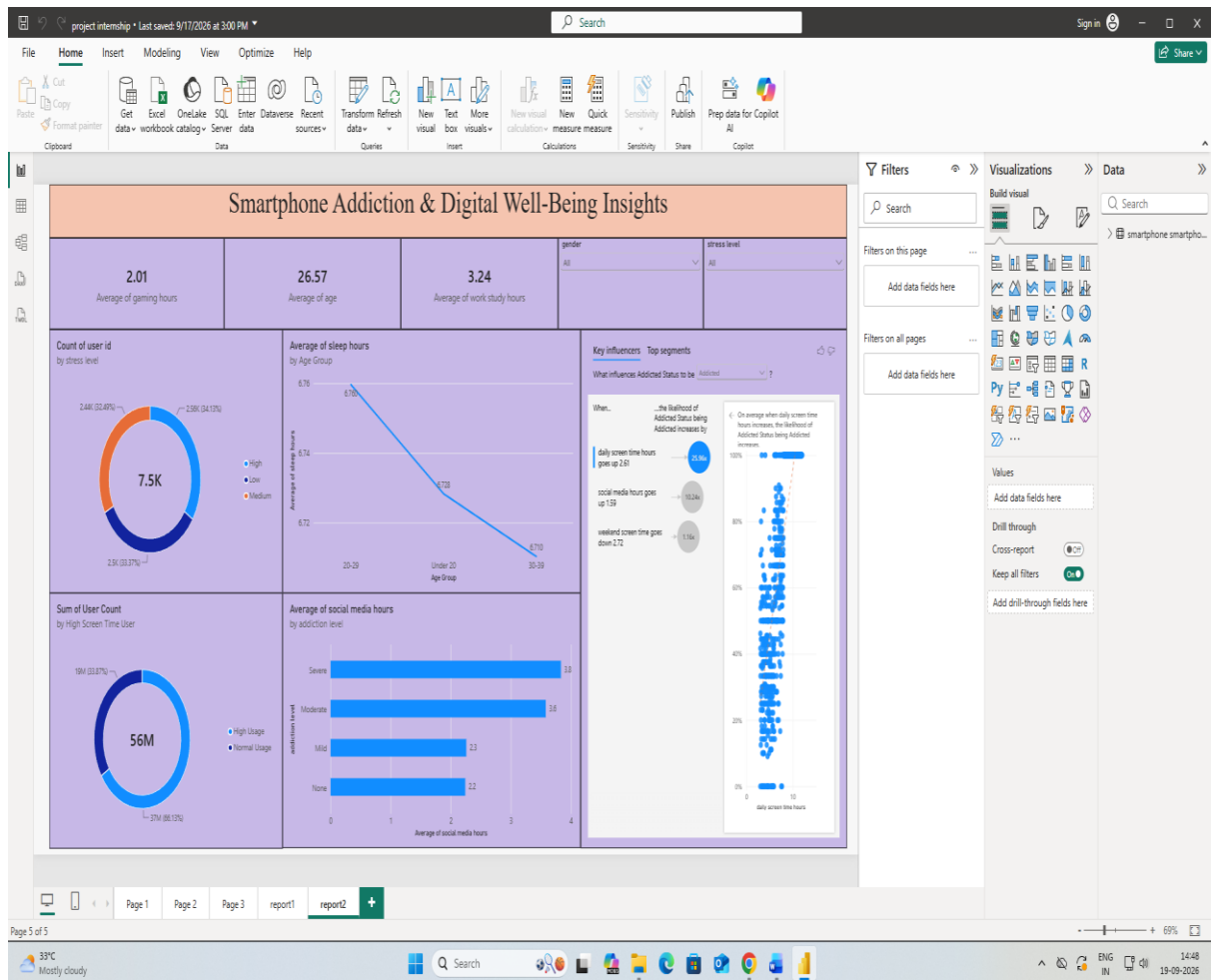
- Severe: **2.43K (32.45%)**
- Mild: **1.37K (18.31%)**
- None: **0.82K (10.92%)**

The dashboard also provides filters for addiction level, age group, gender and stress level.

6. Report

6.1 Report Design Screenshot with Insights





Important Report Insights

Addiction Status

- Addicted Users: **5,308**
- Addicted percentage: **70.77%**

Age Group Distribution

- 20–29: **3,692 users**
- 30–39: **2,569 users**
- Under 20: **1,239 users**

Stress Level

- High: approximately **2.56K (34.13%)**
- Medium: approximately **2.50K (33.37%)**
- Low: approximately **2.44K (32.49%)**

Social Media Usage by Addiction Level

- Severe: **3.8 hours**
- Moderate: **3.6 hours**
- Mild: **2.3 hours**
- None: **2.2 hours**

Weekend Screen Time by Addiction Level

- Severe: **10.4 hours**
- Moderate: **10.1 hours**
- Mild: **7.3 hours**
- None: **6.2 hours**

7. Performance Testing

7.1 Utilization of Data Filters

Interactive filters/slicers are used to explore the dashboard based on:

- Age Group
- Gender
- Addiction Level
- Stress Level

These filters allow users to view specific segments of the smartphone usage data.

7.2 No. of Calculation Columns / Measures

The Power BI analysis uses calculated fields/measures for important metrics such as:

- User Count
- Average Daily Screen Time
- Average Weekend Screen Time
- Average Sleep
- Average Social Media Usage

- Average Gaming Usage
- Average App Opens
- Average Notifications
- Average Work/Study Time

An **Age Group** calculated field was created for segment-level analysis.

7.3 No. of Visualizations

The report uses:

- KPI Cards
- Donut Charts
- Bar/Column Charts
- Scatter Plot
- Line Chart
- Decomposition Tree
- Key Influencers
- Interactive Filters/Slicers

8. Conclusion / Observation

The **Smartphone Usage, Digital Addiction & Well-Being Analytics** project converts **7,500 smartphone-usage records** into an interactive Power BI report.

The analysis covers:

- Smartphone engagement
- Addiction levels
- Daily and weekend screen time
- Sleep
- Social media
- Gaming
- Stress

- Age groups
- Gender
- App openings
- Notifications
- Work/study indicators

The dashboard makes these measures easier to compare and explore using KPI cards, charts, filters, a decomposition tree and Key Influencers analysis.

The Key Influencers visual identified associations between **Addicted Status** and factors such as daily screen time, social media hours and weekend screen time. These are **associations shown by the Power BI analysis and do not prove causation**.

9. Future Scope

Future development can include:

- Larger datasets for analysis.
- Additional digital well-being indicators.
- More detailed usage analysis.
- Automated/real-time data updates.
- Additional predictive analytics.
- More advanced interactive dashboard features.

10. Appendix

10.1 Source Code (if any)

The project primarily uses:

- **Microsoft Power BI Desktop**
- **Power Query**
- **Smartphone_Usage.csv**

Power Query is used for data preprocessing, while Power BI is used for modelling, measures, analysis and visualization.

Source Code: Not applicable as a separate coding component; the project is developed primarily using Power BI and Power Query.

10.2 GitHub & Project Demo Link

GitHub Link: <https://github.com/Priskilla-Data/Smartphone-Usage-Digital-Addiction-Well-Being-Analytics>

Project Demo Link: _____